№3

LaTeX

2024

, ,

LaTeX (inline display), **amsmath**, **bm**, **mathtools**, : , , .

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-
-
- `align`
-
-

: inline display

: \(... \) \[... \]

, .

```
Transcript written on exercise_3_8.log.
PS C:\Users\sidra\Desktop> notepad exercise_3_8.tex
PS C:\Users\sidra\Desktop> pdflatex exercise_3_8.tex
This is pdfTeX, Version 3.1415926535-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise_3_8.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsmath.sty
For additional information on amsmath, use the `?' option.
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amstext.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsgen.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsbsy.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsopn.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/amssymb.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/amsfonts.sty))
(c:/texlive/2025/texmf-dist/tex/latex/tools/bm.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
(./exercise_3_8.aux) (c:/texlive/2025/texmf-dist/tex/latex/amsfonts/umsa.fd)
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/umsb.fd)
Missing character: There is no b in font msbm10!
```

. 1: inline display

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```
PS C:\Users\sidra\Desktop> notepad exercise_3_8_options.tex
PS C:\Users\sidra\Desktop> pdflatex exercise_3_8.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise_3_8.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsmath.sty
For additional information on amsmath, use the `?' option.
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amstext.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsgen.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsbsy.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsopn.sty))
```

. **2:** PDF

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{} .

```
PS C:\Users\sidra\Desktop> pdflatex exercise_3_8_options.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise_3_8_options.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/leqno.clo)
(c:/texlive/2025/texmf-dist/tex/latex/base/fleqn.clo)
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsmath.sty
For additional information on amsmath, use the `?' option.
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amstext.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsgen.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsbsy.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsopn.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/amssymb.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/amsfonts.sty))
(c:/texlive/2025/texmf-dist/tex/latex/tools/bm.sty)
```

. 3:

\diff

\newcommand.

1 Exercice 3.8 : Mathématiques avancées

1.1 Passage des mathématiques en ligne au display

Formule en ligne : $f(x) = \int_0^x e^{-t^2} dt$

Même formule en display :

$$f(x) = \int_0^x e^{-t^2} dt$$

1.2 Lettres grecques

Lettres minuscules : $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \pi, \rho, \sigma, \tau, \upsilon, \phi, \chi, \psi, \omega$

Lettres majuscules : $\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Upsilon, \Phi, \Psi, \Omega$

1.3 Changements de polices

Polices en mode mathématique :

$$\begin{gathered}
\mathrm{mathrm} : \text{texte romain} \\
\mathbf{mathbf} : \textbf{texte gras} \\
\mathit{mathit} : \textit{texte italique} \\
\mathsf{mathsf} : \text{sans serif} \\
\mathtt{mathtt} : \texttt{monospace} \\
\gtrless \partial \approx \approx : \leqslant \partial \top \bowtie \partial \setminus \ \bowtie \leqslant \\
\Updownarrow \dashv \sqcup (\lfloor \dashv \Updownarrow : \lfloor \dashv \Updownarrow \rfloor) \nabla \dashv \surd \langle \rangle \amalg \sqcap \rceil
\end{gathered}$$

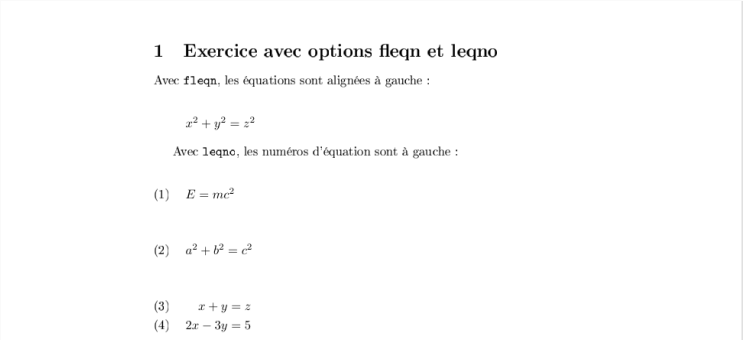
1.4 Test d'imbrication

Test d'imbrication : *texte imbriqu*

align amsmath

align* . :

- & ;
- \\ ;
- \quad ;
- \text{} .



1 Exercice avec options fleqn et leqno

Avec `fleqn`, les équations sont alignées à gauche :

$$x^2 + y^2 = z^2$$

Avec `leqno`, les numéros d'équation sont à gauche :

$$E = mc^2 \tag{1}$$

$$a^2 + b^2 = c^2 \tag{2}$$

$$x + y = z \tag{3}$$

$$2x - 3y = 5 \tag{4}$$

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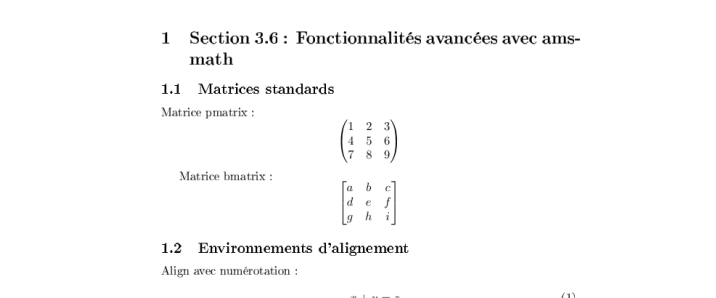
```
PS C:\Users\sidra\Desktop> notepad exercise_3_6_simple.tex
PS C:\Users\sidra\Desktop> notepad exercise_3_7_simple.tex
PS C:\Users\sidra\Desktop> pdflatex exercise_3_6_simple.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise_3_6_simple.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsmath.sty
For additional information on amsmath, use the `?' option.
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amstext.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsgen.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsbsy.sty)
(c:/texlive/2025/texmf-dist/tex/latex/amsmath/amsopn.sty))
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/amssymb.sty
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/amsfonts.sty))
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
```

. 6: PDF

bm :

- `\mathrm{}` — Roman;
- `\mathit{}` — Italic;
- `\mathbf{}` — Bold;
- `\mathbb{}` — Blackboard;
- `\bm{}` — .



1 Section 3.6 : Fonctionnalités avancées avec ams-math

1.1 Matrices standards

Matrice pmatrix :

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Matrice bmatrix :

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

1.2 Environnements d'alignement

Align avec numérotation :

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```
(c:/texlive/2025/texmf-dist/tex/latex/amsfonts/umsb.fd)
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}]
(./exercise_3_7_simple.aux) ) <c:/Users/sidra/.texlive2025/texmf-var/fonts/pk/l
jfour/jknappen/ec/ecrm1000.600pk> <c:/Users/sidra/.texlive2025/texmf-var/fonts/
pk/ljfour/jknappen/ec/ecbx1200.600pk> <c:/Users/sidra/.texlive2025/texmf-var/fo
nts/pk/ljfour/jknappen/ec/ecbx1440.600pk><c:/texlive/2025/texmf-dist/fonts/type
1/public/amsfonts/cm/cmbx10.pfb><c:/texlive/2025/texmf-dist/fonts/type1/public/
amsfonts/cm/cmex10.pfb><c:/texlive/2025/texmf-dist/fonts/type1/public/amsfonts/
cm/cmmi10.pfb><c:/texlive/2025/texmf-dist/fonts/type1/public/amsfonts/cm/cmmi7.
pfb><c:/texlive/2025/texmf-dist/fonts/type1/public/amsfonts/cm/cmr10.pfb><c:/te
xlive/2025/texmf-dist/fonts/type1/public/amsfonts/cm/cmr5.pfb><c:/texlive/2025/
texmf-dist/fonts/type1/public/amsfonts/cm/cmr7.pfb><c:/texlive/2025/texmf-dist/
fonts/type1/public/amsfonts/cm/cmsy10.pfb><c:/texlive/2025/texmf-dist/fonts/typ
e1/public/amsfonts/cm/cmsy7.pfb><c:/texlive/2025/texmf-dist/fonts/type1/public/
amsfonts/cm/cmti10.pfb><c:/texlive/2025/texmf-dist/fonts/type1/public/amsfonts/
symbols/msbm10.pfb>
Output written on exercise_3_7_simple.pdf (1 page, 119773 bytes).
Transcript written on exercise_3_7_simple.log.
PS C:\Users\sidra\Desktop> start exercise_3_6_simple.pdf
PS C:\Users\sidra\Desktop> start exercise_3_7_simple.pdf
```

. **8:** PDF

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- LaTeX;
- **amsmath**, **bm**, **mathtools**;
- `align`;
- ;
- .